

Welcome!



1st Workshop on

Generative AI for Synthetic Data in Learning Analytics

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University of Bergen



**The 16th International Learning Analytics
& Knowledge Conference**

April 27 - May 1, 2026

Bergen, Norway



Team Intro

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UNIVERSITY OF BERGEN

CENTRE FOR THE SCIENCE OF LEARNING & TECHNOLOGY

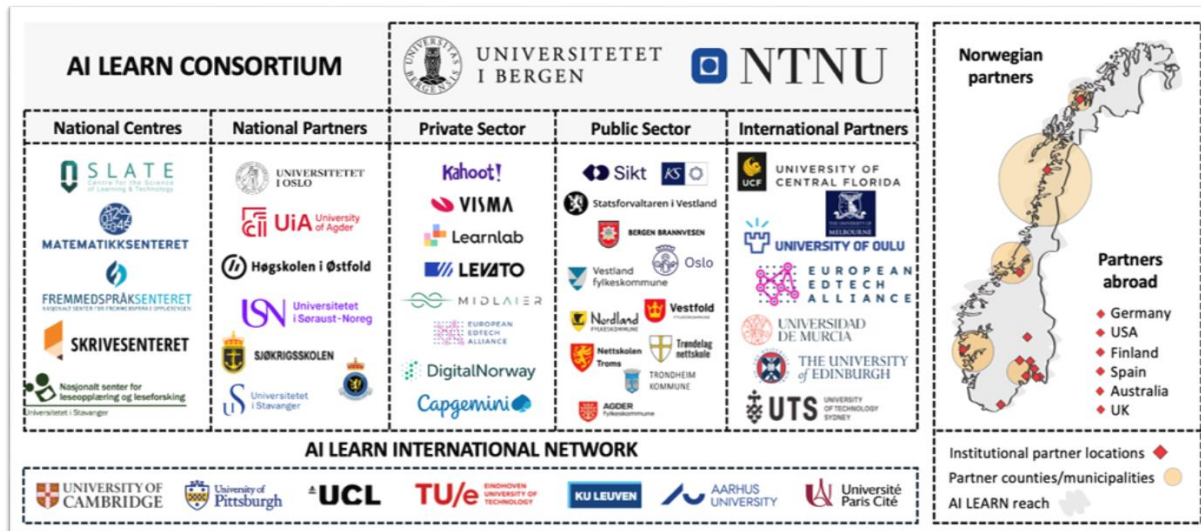
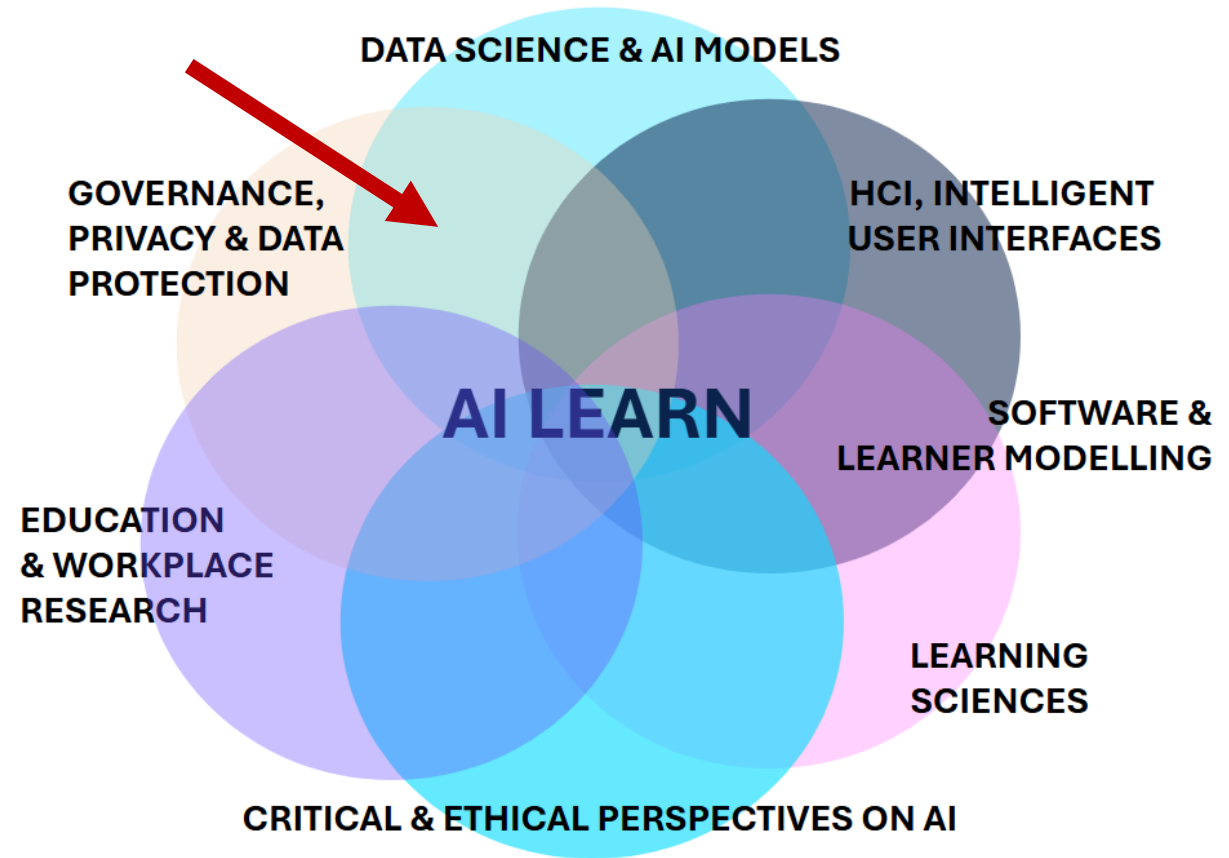
SLATE is a leading centre in the development of knowledge about how educational data can be used to support knowledge, learning, and teaching processes. Drawing on interdisciplinary collaboration we investigate the technological, pedagogical, cultural, ethical, and legal aspects of learning analytics (LA) and artificial intelligence in education (AIEd) and promote the responsible use of technology in education.

SLATE gathers researchers from multiple disciplines, including information science, computer science, cognitive science, sociology, pedagogy and law, thereby conducting integrated research that will advance the frontiers of sciences of learning, as well as inform education practice and policy.

<http://slate.uib.no>

AI LEARN

<http://ailearn.no>



35 partners



Short self-intro about yourself



Agenda

- Booting up...Synthetic data
- What is synthetic data
- Utility and Privacy of synthetic data
- Coffee Break (10.30 – 11.00)
- Guest speaker (Ito Hibiki, Kyoto University- Japan)
- Labs

Dystopia:

Privacy & Data Protection issues

Universities are tracking their students. Is it clever or creepy?

Learning analytics are becoming increasingly popular for improving learning and cutting drop-out rates - but critics question the impact on privacy



▲ 'Some students are concerned about us continuously monitoring in a Big Brother fashion.' Photograph: Alamy

The Data Challenge in Educational Research



Privacy Barriers

GDPR and data protection regulations restrict collection, storage, and sharing of student data across institutions and borders.



Data Scarcity

Predictive models use small samples. Insufficient data limits scalability and generalisability of research.



Quality & Bias

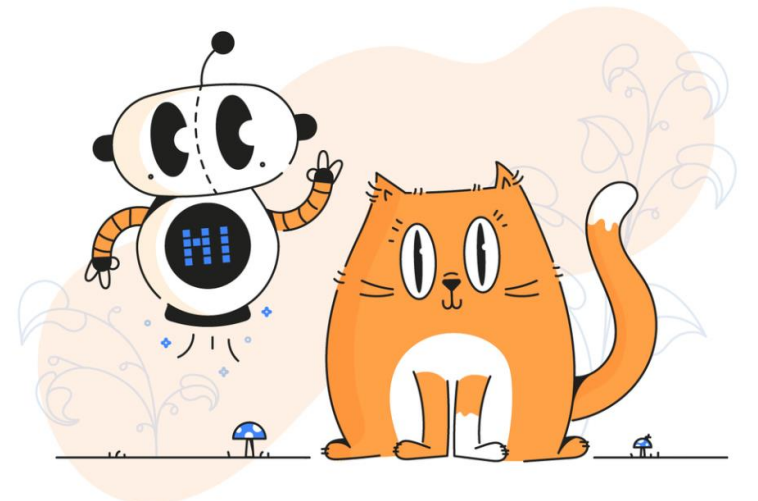
Incomplete datasets, collection biases, and class imbalances undermine the validity of educational models and interventions.

Synthetic data generation offers a promising path to address these interconnected challenges.

gretelTM

Welcome to Gretel!

The developer platform for synthetic data.



gretel

BETA

Search or go to...

+

🔑

🌐

Account

👤

My Gretel

Dashboard

API Key

My Projects

Projects

Telemetry Data

Invites

New Project

Telemetry Data

Telemetry data captured from distributed weather stations.

Models

Data Sources

Members

Settings

Connect

New Model

Run

← Back

quick-brown-fox Completed

Owner

Model type

Quality Score

Training time

Ada Lovelace created 5d ago

Synthetics - Cloud

86 / 100

4m 32s

Data

Activity

Config

Report

quick-brown-fox created by Ada Lovelace August 10, 2021

Processing dataset 1s
data_collection_001.csv (0.54 MB)

Validating training configuration 12s
my_custom_config.yaml is valid.

Starting worker 12s
synthetics@60787d916cc275238501d980 created successfully

Training Synthetic Data Model 12s
Training 100 epochs

Generating new records 12s

Training field overview ?

Field	Unique	Missing	Ave. Length	Type	Distribution Stability
<u>id</u>	5110	0	4.84	Numeric	Excellent
<u>bmi</u>	418	201	4.00	Numeric	Excellent
<u>avg_glucose_level</u>	3979	0	5.29	Numeric	Excellent
<u>smoking_status</u>	4	0	10.08	Categorical	Excellent
<u>age</u>	104	0	3.93	Numeric	Excellent
<u>work_type</u>	5	0	8.25	Categorical	Excellent
<u>gender</u>	3	0	5.17	Categorical	Excellent
<u>heart_disease</u>	2	0	1.00	Categorical	Excellent
<u>Residence_type</u>	2	0	5.00	Categorical	Excellent
<u>ever_married</u>	2	0	2.66	Categorical	Excellent
<u>stroke</u>	2	0	1.00	Categorical	Excellent
<u>hypertension</u>	2	0	1.00	Categorical	Excellent

gretel

Synthetics report

Generated 1/20/2020, 14:35

Your Synthetic Data Quality Score is Excellent

What do these scores mean?



Data Summary Statistics

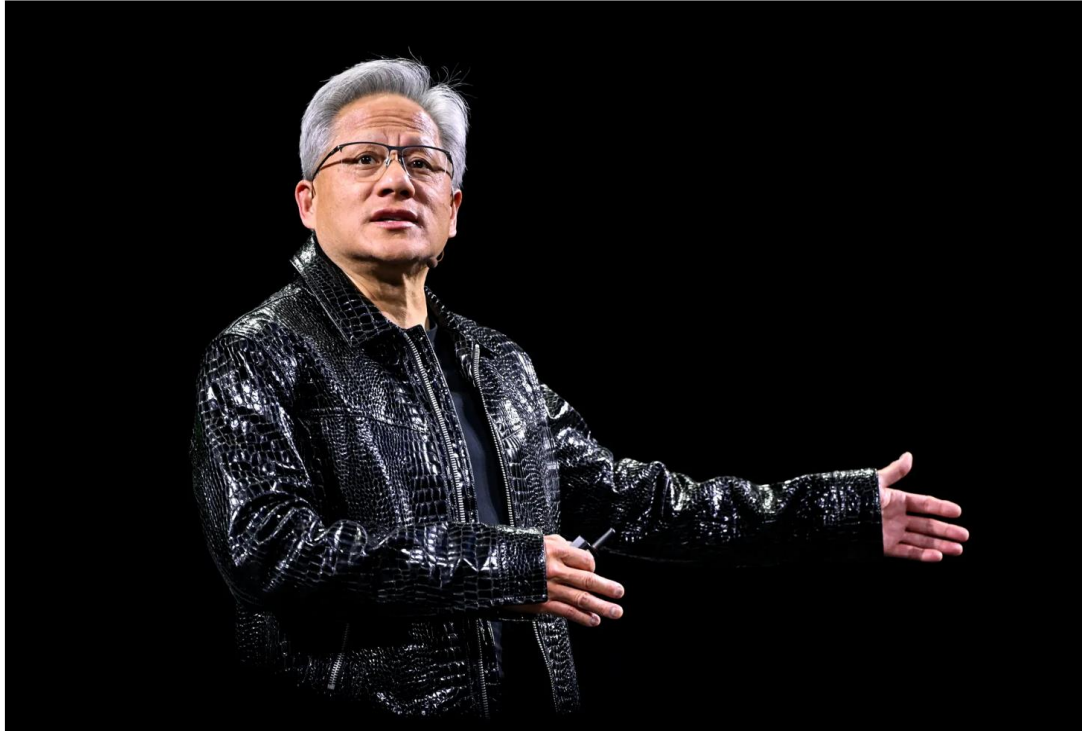


Startup established in 2019

Gretel..raised over \$67 million in capital from notable investors such as Anthos Capital, Greylock, and Moonshots Capital

Nvidia Bets Big on Synthetic Data

Nvidia has acquired synthetic data startup Gretel to bolster the AI training data used by the chip maker's customers and developers.



Nvidia CEO Jensen Huang addresses participants at the keynote of CES 2025 in Las Vegas, Nevada. PHOTOGRAPH: ARTUR WIDAK/GETTY IMAGES

Generate Differentially Private Synthetic Text with Gretel GPT

```
1 models:
2   - gpt_x:
3     ...
4     privacy_params:
5       dp: true
6       epsilon: 8
7       delta: auto
8       entity_column_name: user_id
9
```

Nvidia reportedly acquires Gretel for \$320M+ to strengthen AI training tools



BY DUNCAN RILEY

Report: Nvidia Acquires Synthetic Data Firm Gretel in Nine-Figure Deal



Faizan Farooque
March 19, 2025 • 1 min read

CNET YOUR GUIDE TO A BETTER FUTURE

Nvidia Reportedly Bought a Synthetic Data Firm. So What's Synthetic Data?

Generative AI models must be trained on a lot of information, and sometimes you have to make it up.

Sold for \$320 million

The huge **interest** behind synthetic data

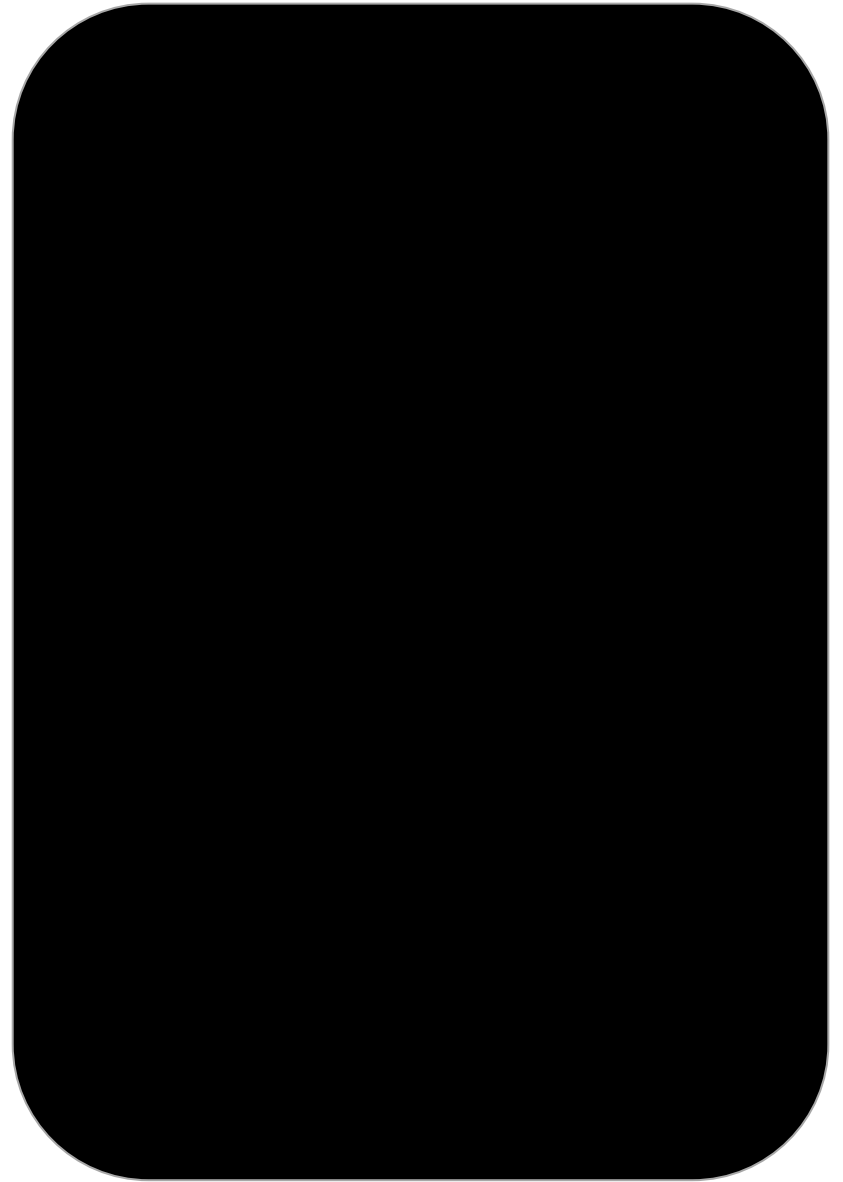
- The **demand** for the large amount of data
 - To train and build AIML models
- The **advancement of methods** to generate data
 - High quality
 - Large volume
- Practical and strong **privacy** approach



Privacy → Anonymisation?

Who is undercoverd?





Different forms of anonymisation

useful
but not private



What is Synthetic Data?

Synthetic data is artificial data designed to mimic real-world data. It's generated through statistical methods or by using artificial intelligence (AI) techniques like Machine Learning, deep learning, and generative AI.

Synthetic Data

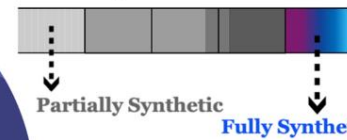
Definition (by the Royal Society and Alan Turing Institute): “data that has been generated using a purpose-built mathematical model or algorithm, with the aim of solving a (set of) data science task(s)”.

How are synthetic data generated?



GANs
VAEs

How are synthetic data classified?



Partially Synthetic

Fully Synthetic

DIGITAL TWINS

Virtual replicas of physical systems or processes that can be used to simulate and predict their behavior in real-time.

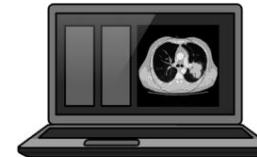


BIAS & QUALITY

Lack of robust method to audit the perpetuation of bias, accuracy and representativeness or real-world medical scenario.

DATA TYPE

Synthetic data can enrich the volume and diversity of datasets including tabular data and imaging alone or combined.



PRIVACY CONCERNS



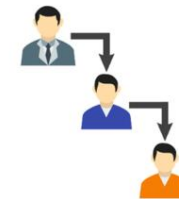
1 Regulatory Agencies

GDPR and HIPAA are not sufficient or up-to-date to cover possible leakage of patients' information from synthetic dataset.



2 Differential Privacy

Based on a mathematical constraint that adds noise to the original dataset to protect individuals privacy.



3 Chain of Custody

To ensure the integrity, security, and privacy of data throughout its lifecycle (data sharing, storing, and disposal).

Giuffrè, M., & Shung, D. L. (2023). Harnessing the power of synthetic data in healthcare: innovation, application, and privacy. NPJ digital medicine, 6(1), 186.

Why Use Synthetic Data?



ADDRESS DATA SCARCITY
IN SENSITIVE DOMAINS
(E.G., HEALTHCARE,
FINANCE)



ENHANCE PRIVACY BY
AVOIDING THE USE OF
REAL PERSONAL DATA



IMPROVE TRAINING FOR
MACHINE LEARNING
MODELS WITH DIVERSE
DATASETS



MITIGATE BIAS AND
ENSURE FAIRNESS



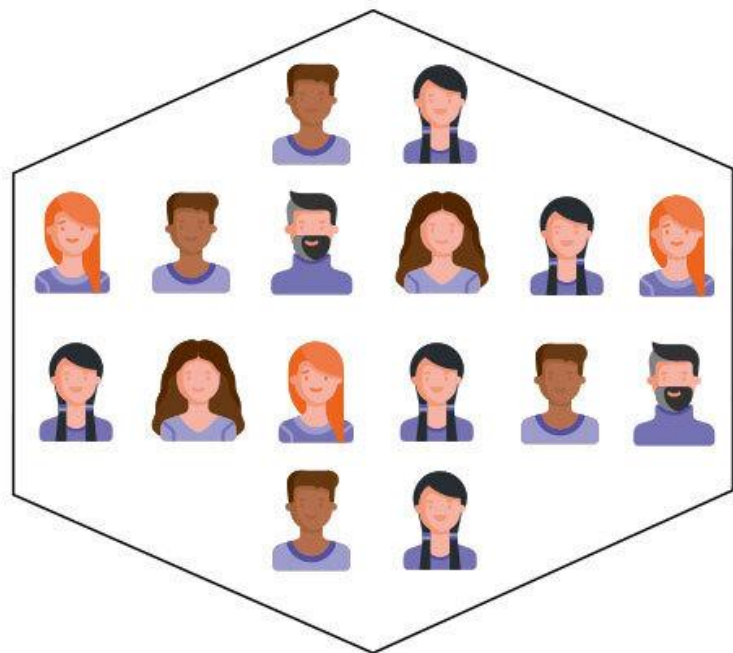
ENABLE TESTING IN SAFE
AND CONTROLLED
ENVIRONMENTS

01

Synthetic Data for Privacy Preservation



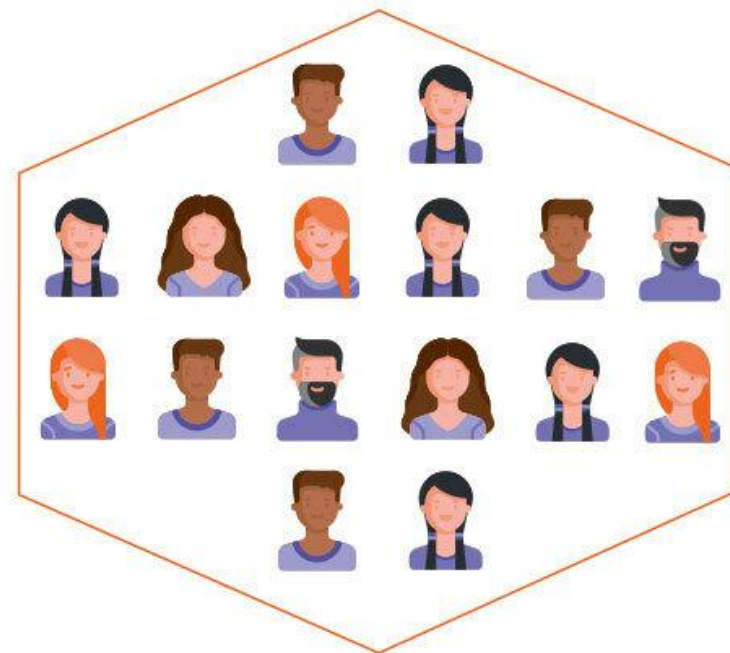
Original Dataset O



**Generative
Model**



Synthetic Dataset S

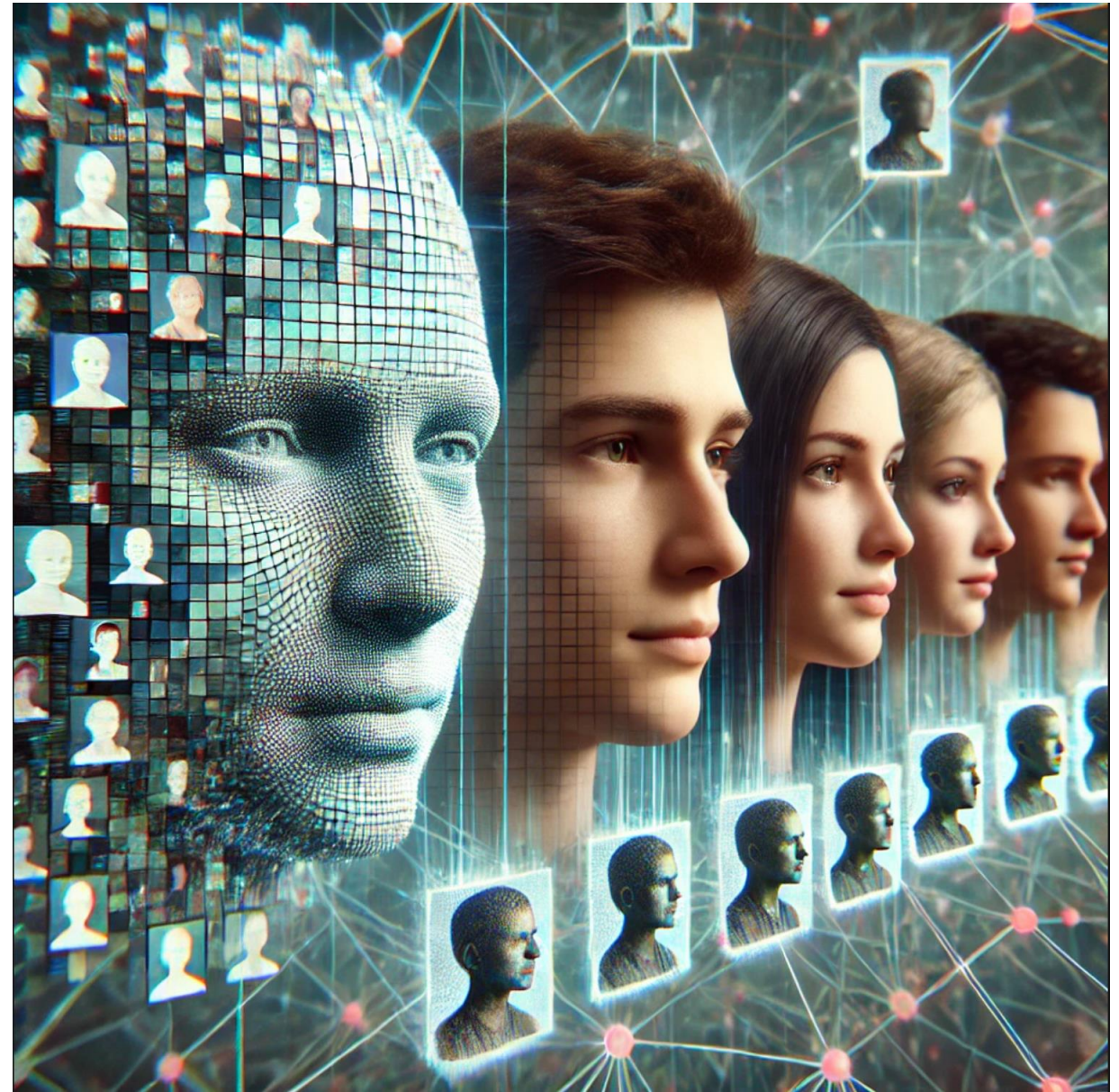


02

Synthetic Data for Performance & Scalability



Artificial Students that resemble actual students?



Comparison of Real and Synthetic Data for Dataset

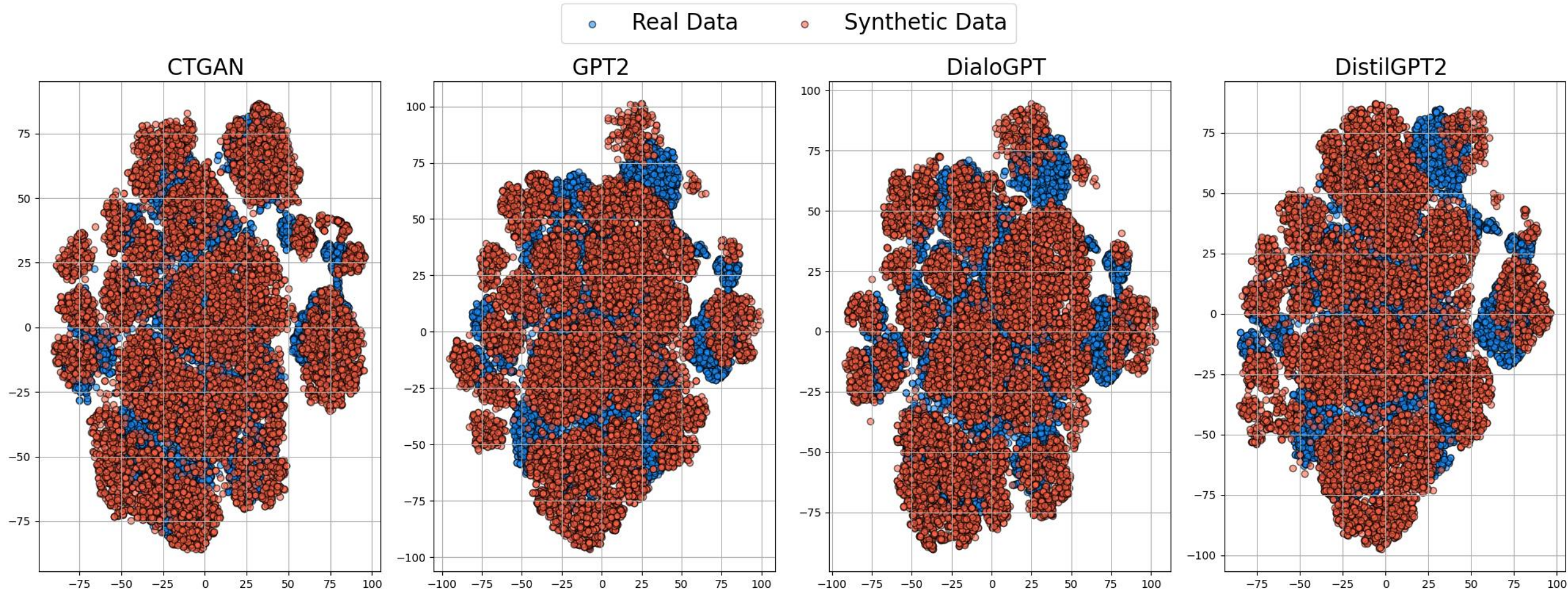


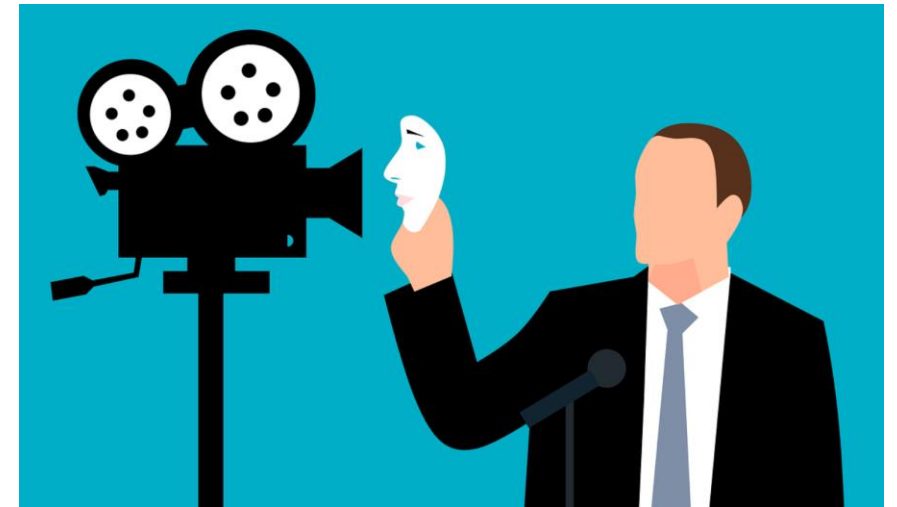
Figure 3: This figure shows the t-SNE projection comparing real and synthetic data generated by different methods for the student info 30% dataset.

93% preserve the structure of the original dataset!

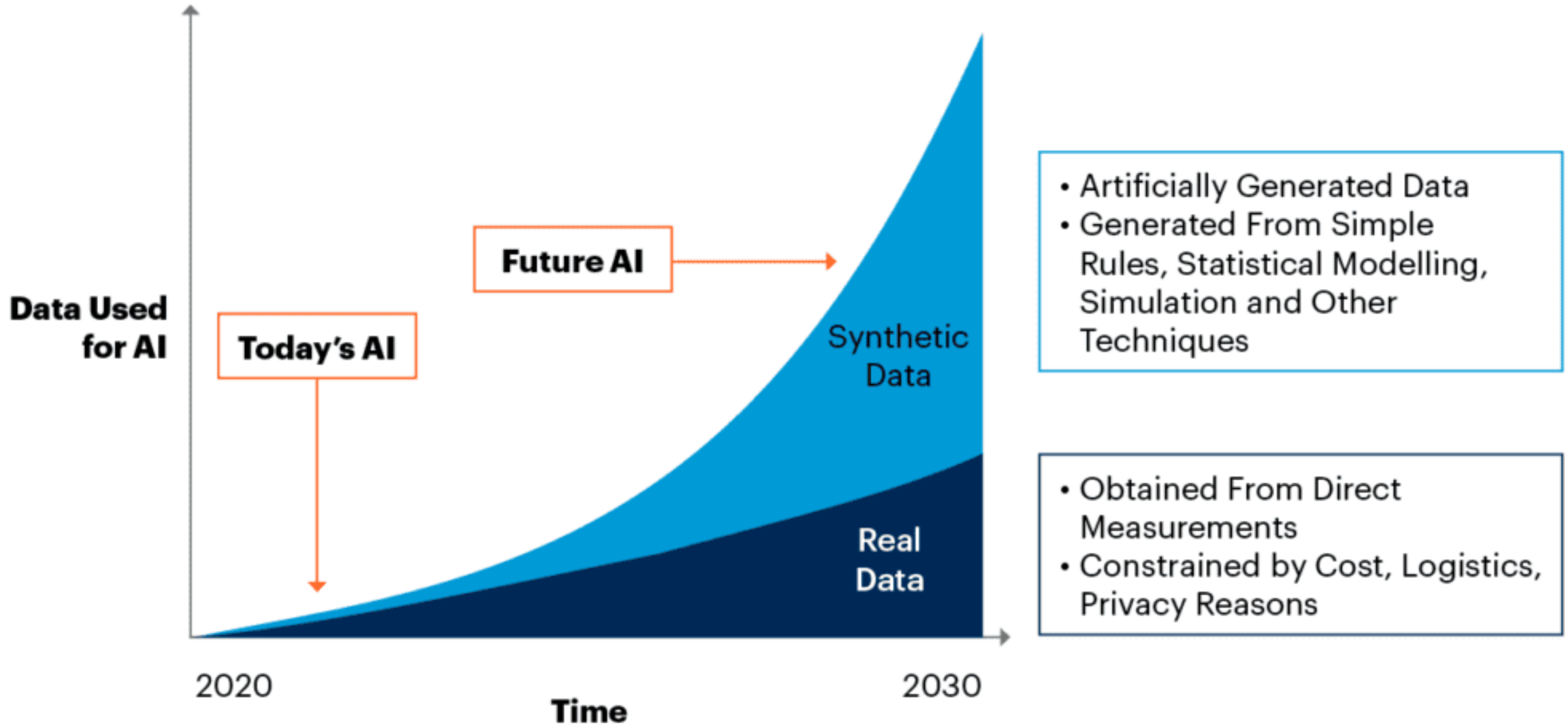
What data is referred to in *synthetic data*
in the context of this Workshop?

Synthetic data vs real data

- Data collected from actual:
 - Events, users, clickstream, sensors, admission
- Challenges:
 - Expensive to collect
 - Expensive to label
 - Bias and imbalance
 - Example: Societal reasons..
 - Availability concerns
 - Ex: Healthcare conditions, special setup...
 - Privacy

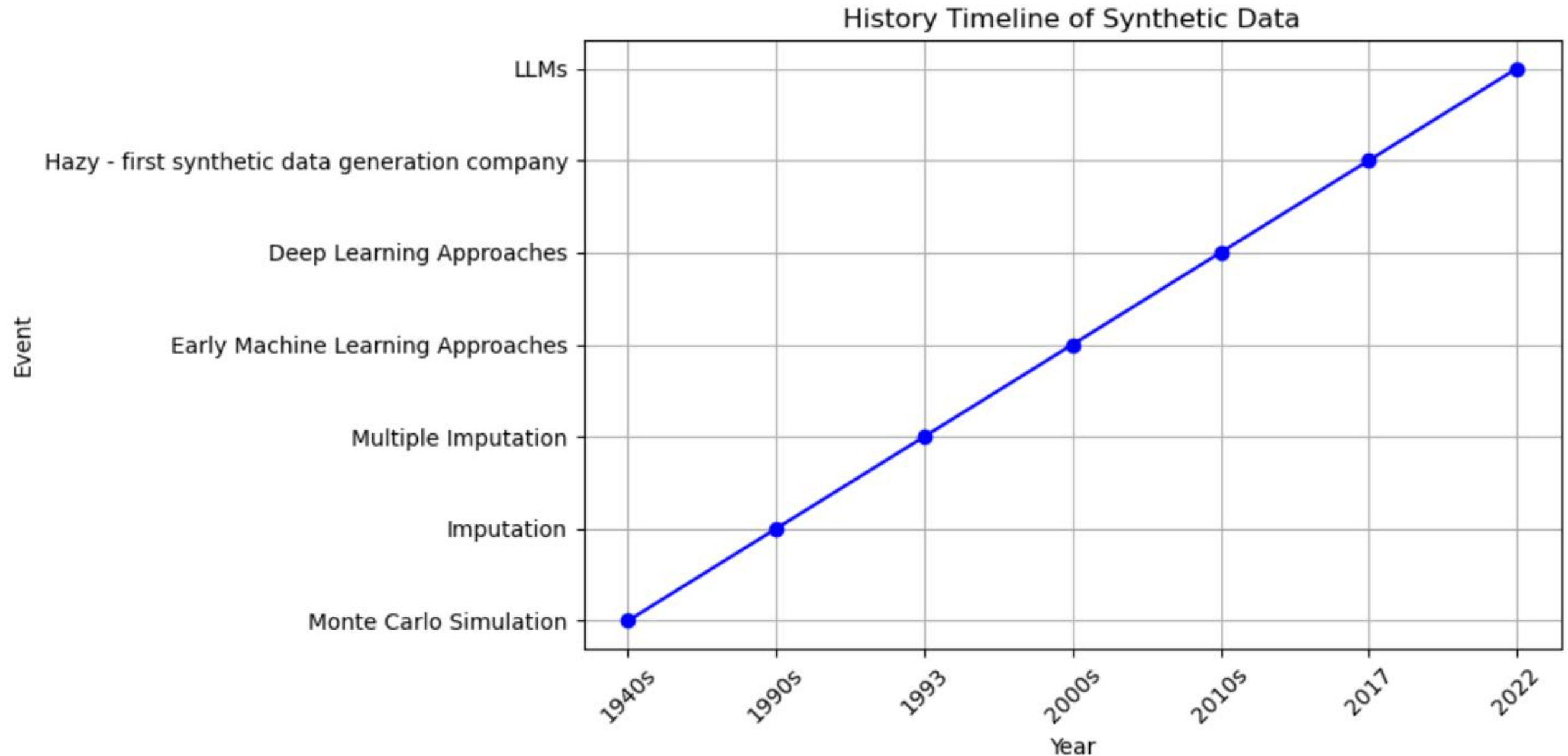


By 2030, Synthetic Data Will Completely Overshadow Real Data in AI Models

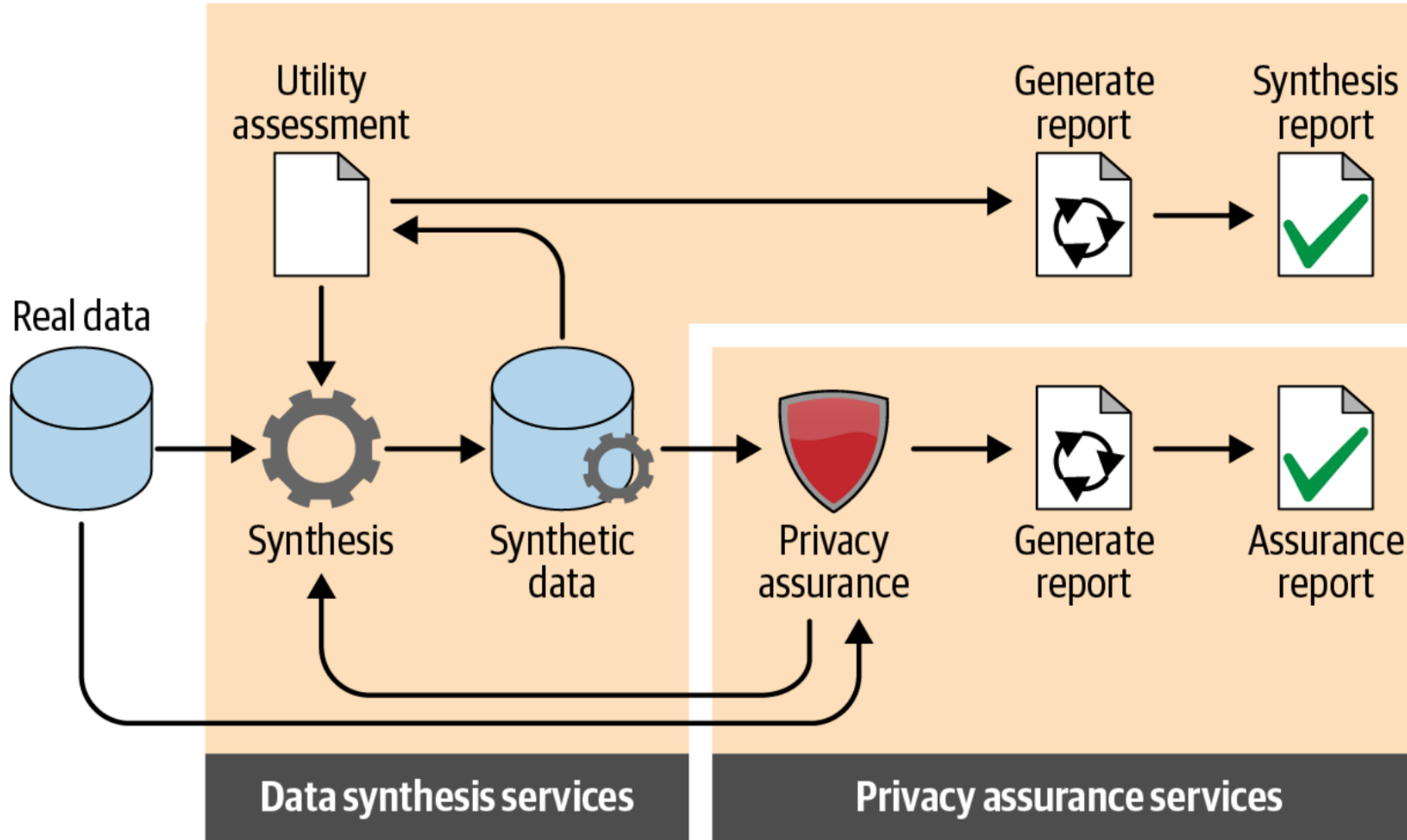


The Spectrum of Synthetic Data

History Recap of Synthetic Data Gen.



How to generate Synthetic Data?



Ethical considerations and data privacy foundations

Ethical Risks in Synthetic Data Use

- False assumption that synthetic = inherently safe
- Potential for re-identification through linkage attacks



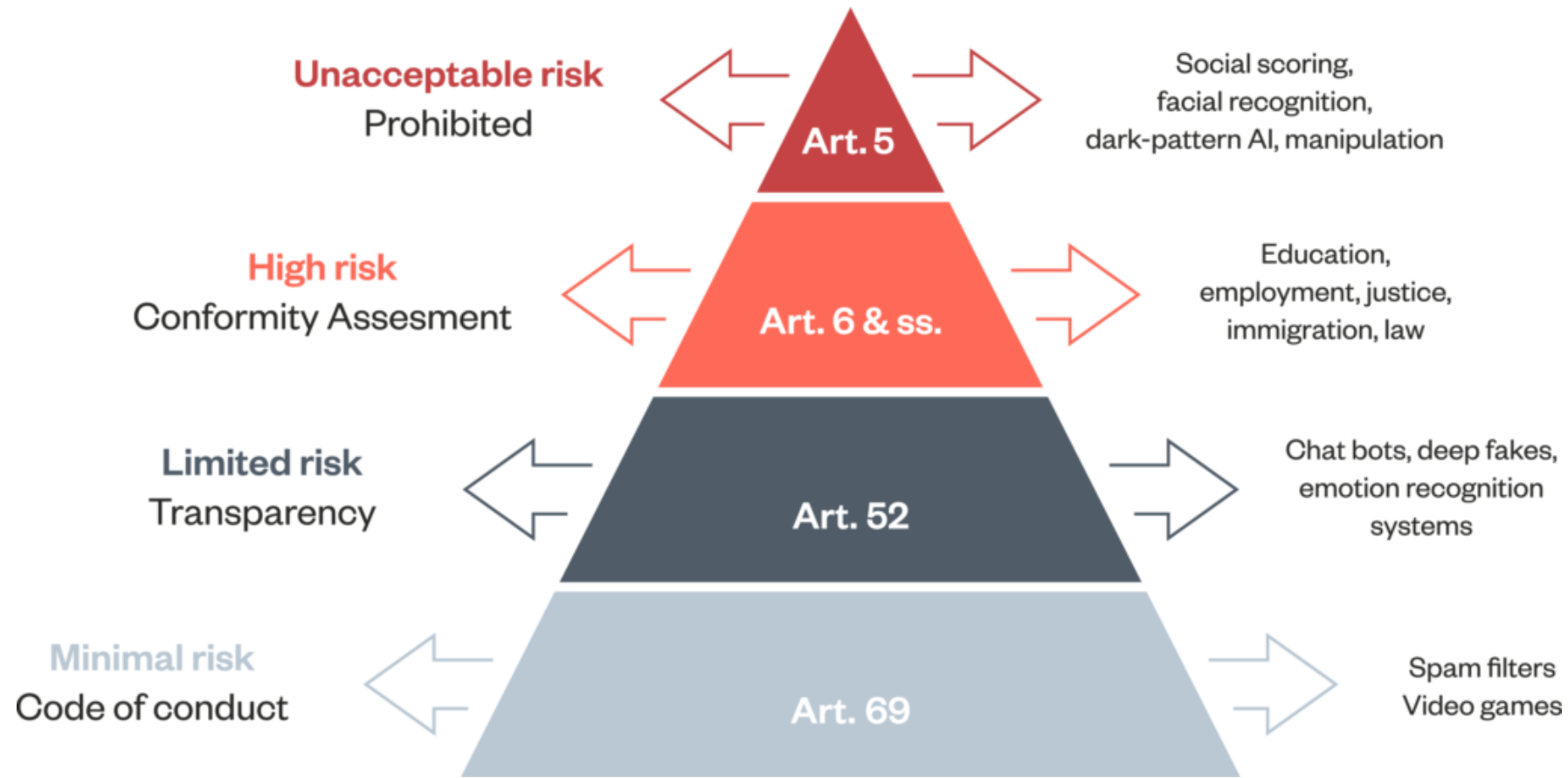
ETHICS

*Synthetic data inherits the ethical burden of the original data **and** introduces new ones!*

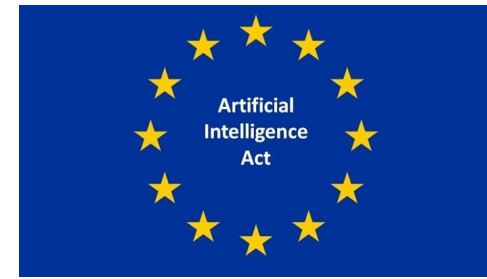
Informed Consent and Transparency

- Many real-world datasets used to train generative models were collected without consent for secondary synthetic use.
- **Best Practices**
 - Obtain broad consent for downstream synthetic uses.
 - Maintain a transparency log documenting how and why synthetic data was generated.
 - Include synthetic data use in public-facing privacy notices.

The AI Act



AI Act (EU)



Synthetic data is mentioned in 3 articles (10, 50, and 59) as well as 2 Recitals (111, 133)

Recital 111

...the capabilities of the model prior to deployment, such as pre-training, **synthetic** data generation and...

Recital 133

A variety of AI systems can generate large quantities of **synthetic** content that becomes increasingly...

Article 10: Data and Data Governance

...and correction cannot be effectively fulfilled by processing other data, including **synthetic** or anonymised data;...

Article 50: Transparency Obligations for Providers and Deployers of Certain AI Systems

...general-purpose AI systems, generating **synthetic** audio, image, video or text content, shall ensure that the...

Article 59: Further Processing of Personal Data for Developing Certain AI Systems in the Public Interest in the AI Regulatory Sandbox

...2 where those requirements cannot effectively be fulfilled by processing anonymised, **synthetic** or other non-personal...



Article 10: Data and Data Governance

5. To the extent that it is strictly necessary for the purpose of ensuring bias detection and correction in relation to the high-risk AI systems in accordance with paragraph (2), points (f) and (g) of this Article, the providers of such systems may exceptionally process special categories of personal data, subject to appropriate safeguards for the fundamental rights and freedoms of natural persons. In addition to the provisions set out in Regulations (EU) 2016/679 and (EU) 2018/1725 and Directive (EU) 2016/680, all the following conditions must be met in order for such processing to occur:

- (a) the bias detection and correction cannot be effectively fulfilled by processing other data, including synthetic or anonymised data;

Bias detection and correction is a key requirement for high-risk AI systems to ensure fairness and non-discrimination.

To protect privacy, developers are encouraged to use synthetic or anonymised data.

However, if using anonymized/synthetic data is not sufficient to detect or correct bias — for example, because anonymisation removes critical demographic indicators — then real personal data (including sensitive data) may be used, but only under strict safeguards.



Article 50: Transparency Obligations

2. Providers of AI systems, including general-purpose AI systems, generating synthetic audio, image, video or text content, shall ensure that the outputs of the AI system are marked in a machine-readable format and detectable as artificially generated or manipulated. Providers shall ensure their technical solutions are effective, interoperable, robust and reliable as far as this is technically feasible, taking into account the specificities and limitations of various types of content, the costs of implementation and the generally acknowledged state of the art, as may be reflected in relevant technical standards. This obligation shall not apply to the extent the AI systems perform an assistive function for standard editing or do not substantially alter the input data provided by the deployer or the semantics thereof, or where authorised by law to detect, prevent, investigate or prosecute criminal offences.

Related: Recital 133

Implication

If synthetic data is **shared externally** (e.g., for research, benchmarking, or public datasets), it *may need to be* **clearly labeled** as artificially generated.

Data usage hierarchy



Necessary data collection



Non-personal data



Anonymised or synthetic



Personal data (only if necessary and under strict guidelines)

GDPR and Synthetic Data



- General Data Protection Regulation
 - Implemented in 2018
- The GDPR does not explicitly mention synthetic data in its articles
- Synthetic data may apply to Recital 26:

The principles of data protection do not apply to anonymous information... This Regulation does not therefore concern the processing of such anonymous information, including for statistical or research purposes

Synthetic data protection: Towards a paradigm change in data regulation?

Ana Beduschi¹ 

Big Data & Society
January–March: 1–5
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Major takeaways

If synthetic data is *truly* anonymous (meaning individuals cannot be identified directly or indirectly), then GDPR does not apply.

If synthetic data can be linked back to real individuals (e.g., via model inversion, membership inference attacks), then it falls under GDPR as it is considered personal data.

You must have a lawful basis (e.g., consent, legitimate interest).